



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Main)

UNIT TEST

04-07-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, PHASE-I LIVE & PHASE-I(A) LIVE]

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on this Test Booklet.

Paper : Physics, Chemistry & Mathematics

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2022

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PHYSICS

SEC-A : (Maximum Marks: 80)

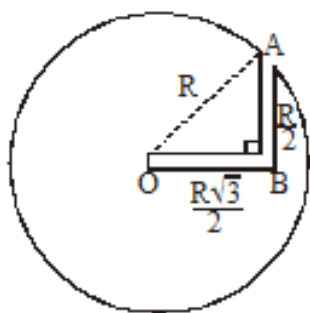
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. One L shaped smooth tunnel is made inside earth. A ball of mass m is released from A. Find speed with which ball strike the surface at B. (g is acceleration due to gravity at surface)

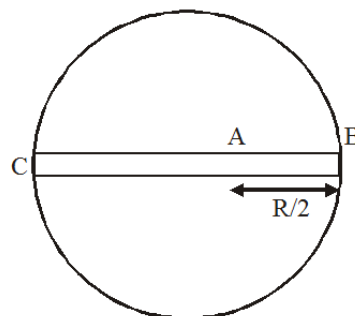


- (A) $\frac{\sqrt{gR}}{2}$
 (B) $\frac{\sqrt{3gR}}{2}$
 (C) $\sqrt{\frac{gR}{2}}$
 (D) $\sqrt{\frac{3gR}{2}}$

2. Six stars of equal mass m moving about the centre of mass of the system such that they are always on the vertices of a regular hexagon of side length a . Find net force on star due to other five stars.

- (A) $\frac{GM^2}{a^2} \left(\frac{5}{4} \right)$
 (B) $\frac{GM^2}{a^2} \left(\frac{5}{4} + \frac{1}{2} \right)$
 (C) $\frac{GM^2}{a^2} \left(\frac{5}{4} + \frac{1}{\sqrt{3}} \right)$
 (D) $\frac{GM^2}{a^2} \left(\frac{5}{4} + \frac{\sqrt{3}}{2} \right)$

3. Two frictionless tunnels AB and AC are dug inside the earth as shown in the figure. A is at $R/2$ distance from centre of earth along the line connecting centre and B. A particle of mass m_1 is projected along AB and another particle of mass m_2 is projected along AC such that both escape. Then ratio of their minimum velocities, so that particle escapes the gravitational field of earth, is



- (A) 1 : 1
 (B) 1 : 2
 (C) 2 : 1
 (D) 3 : 1

4. A uniform wire of ohmic material length of 2m is stretched till its length increased by 10%. The percentage change in resistance to nearest integer is :-

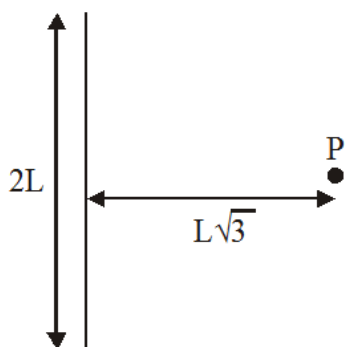
(A) 10 (B) 9 (C) 21 (D) 20

5. A current through a wire depends on time as $i = (4t + 6t^2)A$. Find charge crossed through a section in 5 second in C :-

(A) 170 (B) 300
(C) 200 (D) 500

6. Find the electric field at point P on the perpendicular bisector of a uniformly charged thin wire of length 2L carrying a charge Q. The distance of the point P from the center of rod is

$$L\sqrt{3} :- \left(k = \frac{1}{4\pi\epsilon_0} \right)$$

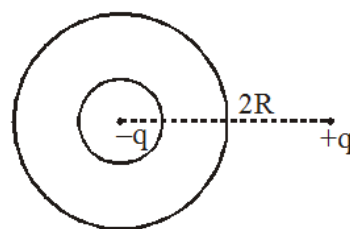


- (A) $\frac{kQ}{L^2\sqrt{3}}$
(B) $\frac{kQ}{2L^2\sqrt{3}}$
(C) $\frac{kQ}{L^2}$
(D) $\frac{kQ}{2L^2}$

7. A solid non conducting sphere having uniform charge Q and radius R. Find self energy of solid sphere. $\left(k = \frac{1}{4\pi\epsilon_0} \right)$

- (A) $\frac{2}{5} \frac{kQ^2}{R}$
(B) $\frac{kQ^2}{2R}$
(C) $\frac{kQ^2}{R}$
(D) $\frac{3}{5} \frac{kQ^2}{R}$

8. A conducting sphere (of radius R) has concentric spherical cavity (radius $\frac{R}{2}$) with a charge $(-q)$ at its centre. A charge $(+q)$ is placed at a distance 2R from the center. Mark the incorrect option :-

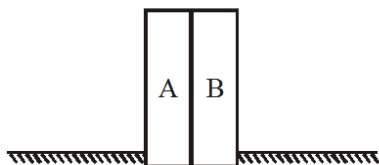


- (A) Charge density on the outer surface is not uniform
(B) Charge density on the inner surface is uniform.
(C) Electric field at the centre due to point charge $(+q)$ is $\frac{1}{4\pi\epsilon_0} \frac{q^2}{R^2}$ in magnitude.
(D) Potential difference between inner and outer surface of sphere is zero.

9. The temperature at the junction of two insulating sheets, having thermal resistance R_1 & R_2 if temperature at the outer ends are T_1 & T_2 respectively :-

(A) $\frac{T_1 R_1 + T_2 R_2}{T_1 + T_2}$ (B) $\frac{T_1 R_1 + T_2 R_2}{R_1 + R_2}$
 (C) $\frac{T_1 R_2 + T_2 R_1}{T_1 + T_2}$ (D) $\frac{T_1 R_2 + T_2 R_1}{R_1 + R_2}$

10. A bimetallic strip consist of metal A & B. If $\alpha_A > \alpha_B$, then on decreasing temperature strip will (α is coefficient of expansion)



- (A) Bend towards A (B) Bend towards B
 (C) Remain straight (D) None of these
11. When temperature of an isotopic metal wire is increased its length increased by 0.01%. Find % decrease in density of material :-
- (A) 0.02% (B) 0.01%
 (C) 0.03% (D) 0.06%
12. 1g ice at 0°C with 1g steam at 100°C mix in a vessel having negligible heat capacity then final composition at equilibrium is (assume no heat loss with surrounding, $L_f = 80 \text{ cal/g}$ and $L_v = 540 \text{ cal/g}$):-
- (A) $\frac{2}{3}\text{g ice}$ (B) $\frac{4}{3}\text{g ice}$
 (C) $\frac{2}{3}\text{g steam}$ (D) $\frac{4}{3}\text{g steam}$

13. Your friend is having eye sight problem. She is not able to see clearly a distant uniform window mesh and it appears to her as non-uniform and distorted. The doctor diagnosed the problem as :-

- (A) Astigmatism
 (B) Myopia with astigmatism
 (C) Presbyopia with Astigmatism
 (D) Myopia and hypermetropia

14. An object of mass m is suspended at the end of a massless wire of length L and cross section A . Young modulus of the material of the wire is Y . If the mass is pulled down slightly its frequency of oscillation along the vertical direction :-

(A) $f = \frac{1}{2\pi} \sqrt{\frac{YA}{mL}}$ (B) $f = \frac{1}{2\pi} \sqrt{\frac{YL}{mA}}$
 (C) $f = \frac{1}{2\pi} \sqrt{\frac{mA}{YL}}$ (D) $f = \frac{1}{2\pi} \sqrt{\frac{mL}{YA}}$

15. The incident ray, reflected ray and the outward down normal are denoted by the unit vector $\vec{a}$, $\vec{b}$ & $\vec{c}$ respectively then which of the following is correct :-

(A) $\vec{b} = \vec{a} + 2\vec{c}$ (B) $\vec{b} = 2\vec{a} + \vec{c}$
 (C) $\vec{b} = \vec{a} - 2(\vec{a} \cdot \vec{c})\vec{c}$ (D) $\vec{b} = \vec{a} - \vec{c}$

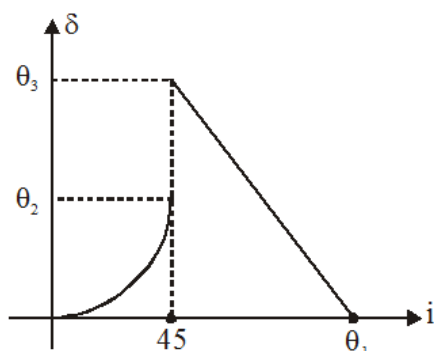
16. 2 particles A & B of equal masses are suspended from two massless springs of constant k_1 & k_2 respectively. If the maximum velocities during oscillations are equal, the ratio of the amplitude of A & B is :-

(A) $\frac{k_2}{k_1}$ (B) $\frac{k_1}{k_2}$ (C) $\sqrt{\frac{k_1}{k_2}}$ (D) $\sqrt{\frac{k_2}{k_1}}$

17. Find out the surface charge density of infinite thin metallic sheet parallel to yz plane passing through $x = 8$. If net field at $x = 3$ is zero due both sheet and infinite line charge of 8 nC/m lying along the z -axis :-

- (A) 0.62 nC/m^2 (B) 0.52 nC/m^2
(C) 0.42 nC/m^2 (D) 0.32 nC/m^2

18. Figure shows graph of angle of deviation δ v/s angle of incidence for a light ray. Incident ray goes from medium 1 (μ_1) to medium 2 (μ_2).



- (a) $\theta_1 = 90^\circ$
(b) Critical angle 45°
(c) $\mu_1 < \mu_2$
(d) $\theta_2 = 60$
(e) $\theta_3 = 90$

Correct set is :-

- (A) abcd (B) abe (C) abde (D) All

19. Dimensional formula for thermal conductivity

$M \rightarrow$ denotes mass

$T \rightarrow$ denotes time

$L \rightarrow$ denotes length

$K \rightarrow$ denotes temperature

- (A) $MLT^{-3}K$ (B) $MLT^{-2}K$
(C) $MLT^{-2}K^{-2}$ (D) $MLT^{-3}K^{-1}$

20. The function of time representing a simple harmonic motion with a period of $\frac{\pi}{\omega}$ is :-

- (A) $\sin \omega t + \cos \omega t$
(B) $\cos \omega t + \cos 2\omega t + \cos 3\omega t$
(C) $\sin 3\omega t$
(D) $3 \cos \left(\frac{\pi}{4} - 2\omega t \right)$

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

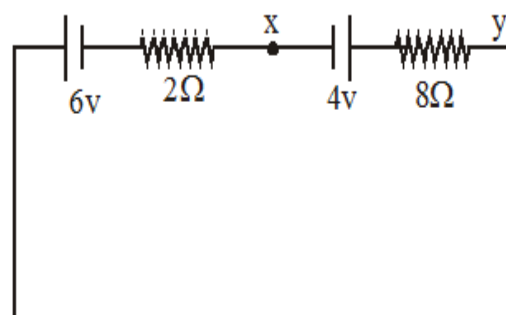
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

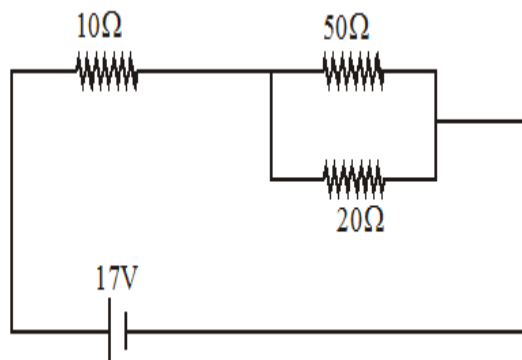
Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. As shown in the figure find potential difference across x and y in (volt).



2. The voltage across the $10\ \Omega$ resistor in the given circuit is ____ volt



3. An object is located at 5 km beneath the water surface. If fractional change in volume is 2%, if ratio of volumetric stress with volumetric strain is $x \times 10^9\ \text{N/m}^2$ find x .
4. A tunnel is dug along a chord of earth having length $\sqrt{3}R$ where R is radius of Earth. A small block is released in the tunnel from the surface of Earth. The particle comes to rest at the centre of tunnel. Find coefficient of friction between the block and the surface of tunnel. Ignore the effect of rotation of Earth. ($\sqrt{3} = 1.74$)
5. A metal cube undergoes pressure of 6 GPa. The percentage change in the length of side of cube is ($B = 10 \times 10^{10}\ \text{Pa}$)

6. Time period of a simple pendulum is T . The time taken to complete $\frac{5}{8}$ oscillations starting from m.p. is $\frac{xT}{y}$ the value of $x + y$.
7. The energy dissipated by a resistor is 10 mJ in 2s when an electric current of 2mA flows through it. The resistance is $x \times 100\ \Omega$. find x .
8. A ray of light passing from air through an equilateral glass prism undergoes minimum deviation when the angle of incidence is $\frac{3}{4}$ th of the angle of prism. Find refractive index of material of prism.
9. If the time period of a two meter long simple pendulum is 2s, the acceleration due to gravity at the place where pendulum is executing SHM is $m\pi^n$, find $m + n$.
10. A deviation of 2° is produced in the yellow ray when prism of crown and flint glass are achromatically combined. Taking dispersive powers of crown and flint glass are 0.02 and 0.03 respectively and refractive index for yellow light for these glasses are 1.5 and 1.6 respectively. The refracting angles for crown glass prism will be ____ $^\circ$ (in degree).

CHEMISTRY

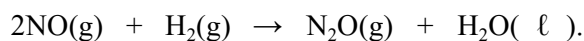
SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

- How many geometrical isomers and stereoisomers are possible for $[\text{Pt}(\text{Br})(\text{Cl})(\text{I})(\text{NO}_2)(\text{NH}_3)(\text{Py})]$
 (A) 3 and 15
 (B) 15 and 30
 (C) 4 and 15
 (D) 4 and 30
- Which of the following complex shows ionization isomerism.
 (A) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$
 (B) $[\text{Cr}(\text{en})_2]\text{Cl}_2$
 (C) $[\text{Cr}(\text{en})_3]\text{Cl}_3$
 (D) $[\text{CoBr}(\text{NH}_3)_5]\text{SO}_4$
- The complex ions $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Fe}(\text{CN})_6]^{4-}$
 (A) Are both octahedral and paramagnetic.
 (B) Are both octahedral and diamagnetic.
 (C) Having same structure but different magnetic character.
 (D) Have different structures but same magnetic character.

- The formula of Diamminetetrachloroplatinum (IV), is
 (A) $[\text{Pt}(\text{NH}_3)_2]\text{Cl}_4$
 (B) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$
 (C) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]\text{Cl}_2$
 (D) $\text{K}_4[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$
- Which of the following complex is optically active ?
 (A) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
 (B) $[\text{Co}(\text{en})_3]\text{Br}_3$
 (C) $[\text{PtCl}_4]^{-2}$
 (D) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$
- For which of the following complexes, central atom does not use d-orbital(s) for hybridisation ?
 (A) $[\text{Ni}(\text{CN})_4]^{4-}$
 (B) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 (C) $[\text{PtCl}_4]^{2-}$
 (D) None of these
- Considering the elements fluorine, chlorine, oxygen and sulphur, the correct order of their electron affinity values is
 (A) $\text{F} > \text{Cl} > \text{O} > \text{S}$
 (B) $\text{F} > \text{O} > \text{Cl} > \text{S}$
 (C) $\text{Cl} > \text{F} > \text{S} > \text{O}$
 (D) $\text{O} > \text{F} > \text{S} > \text{Cl}$

8. For the reaction



Experiments were done to determine rate law expression.

Experiment 1: The values of $-\frac{dp}{dt}$ (rate of reaction in terms of pressure change per unit time), obtained by experiment were 2.25 torr/sec and 0.25 torr/sec, when pressures of NO(g) were 450 torr and 150 torr respectively. The pressure of H₂(g) being constant in given experiment.

Experiment 2: The values of $-\frac{dp}{dt}$ (rate of reaction in terms of pressure change per unit time), obtained by experiment were 1.90 torr/sec and 0.95 torr/sec, when pressures of H₂(g) were 290 torr and 145 torr respectively. The pressure of NO(g) being constant in given experiment.

Which analytical statement is correct on the basis of above experiment(s)

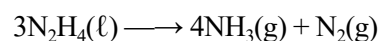
- (I) Order of reaction is 3
- (II) Order with respect to NO(g) and H₂(g) are 3 and 0 respectively
- (III) Order with respect to H₂(g) and NO(g) are 1 and 2 respectively
- (IV) Rate of reaction depend upon the concentration/pressure of reactant(s)

- (A) I, II and IV
- (B) I and III
- (C) I, III and IV
- (D) I only

9. The only **INCORRECT** information is

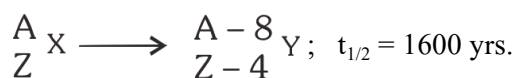
- (A) In isothermal expansion of an ideal gas, internal energy of the gas remains unchanged.
- (B) In adiabatic free expansion of a ideal gas, the internal energy of gas remains unchanged.
- (C) In adiabatic free expansion of an ideal gas, the initial and final temperature of the gas remains same.
- (D) In isobaric expansion of a ideal gas, the initial and final temperature of gas remains same.

10. Calculate the work done in Joules when 1.0 mole of N₂H₄ decomposed completely against a pressure of 1.0 atm at 27°C, (Take R = 8.3 J / K mole)



- (A) -4980
- (B) -14940
- (C) -2490
- (D) -4150

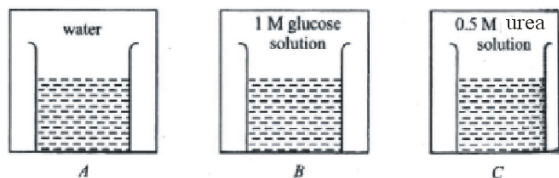
11. For the nuclear reaction ;



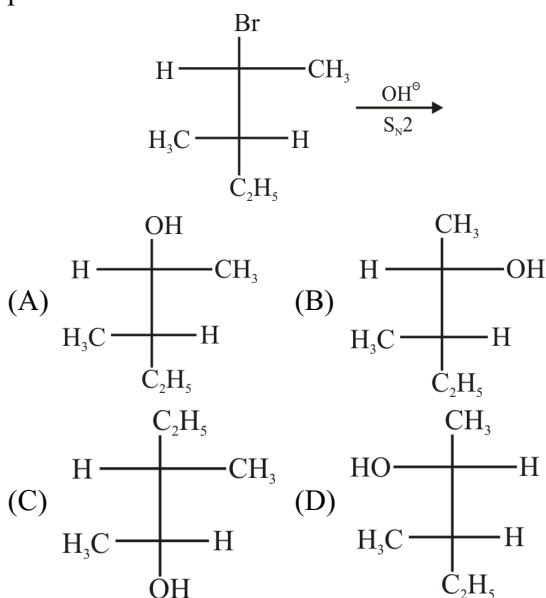
If initial activity of sample of 'X' was 10⁷ dps, how many α-particles will be emitted per second after 4800 years ?

- (A) 1.25 × 10⁶ s⁻¹
- (B) 2.5 × 10⁶ s⁻¹
- (C) 1.25 × 10⁷ s⁻¹
- (D) 2.5 × 10⁷ s⁻¹

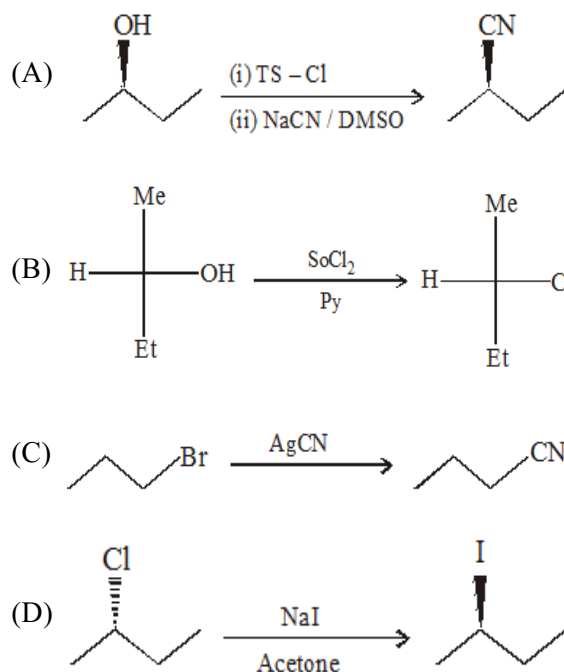
12. In three beakers labelled as (A), (B) and (C), 100 mL of water, 100 mL of 1M aqueous solution of glucose and 100 mL of 0.5 M aqueous solution of urea are taken respectively and kept at same temperature. Which of the following statements is correct ?



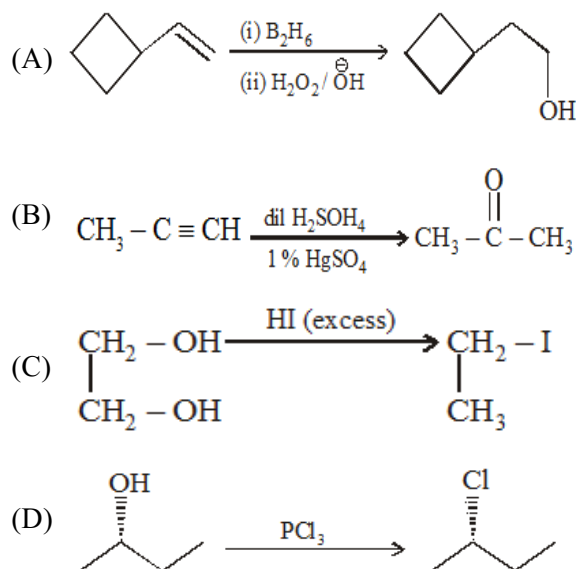
- (A) Vapour pressure in all the three beakers is same.
 (B) Vapour pressure of beaker B is highest among all.
 (C) Vapour pressure of beaker C is highest among all.
 (D) Vapour pressure of beaker B is lower than that of C and vapour pressure of beaker C is lower than that of A.
13. In the following reaction the most probable product will be :



14. Identify reaction correctly matched with their major product?



15. Which reaction is incorrectly matched with its major product?



SEC-B : (Maximum Marks: 40)

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Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. The CFSE for octahedral $[\text{CoCl}_6]^{4-}$ is $18,000 \text{ cm}^{-1}$.

If the CFSE for tetrahedral $[\text{CoCl}_4]^{2-}$ is $x \text{ (in cm}^{-1}\text{)}$.

Find out value of $\frac{x}{100}$

2. Consider the following mineral

Haematite, Magnetite, Limonite, Siderite, Chromite, Wolframite

If no. of minerals having iron as Fe(II) and Fe(III) are x and y respectively. Find the value of x/y

3. Find the total number of complexes are following Sidgwick EAN rule. (Atomic number :

$\text{Mo} \Rightarrow 42, \text{Pt} \Rightarrow 78, \text{Ag} = 47$)

$[\text{Mo}(\sigma\text{-C}_3\text{H}_5)\text{Br}(\text{NH}_3)_2]$, $[\text{Fe}(\eta^5\text{-C}_5\text{H}_5)_2]$,

$[\text{Ni}(\text{DMG})_2]$, $[\text{Mn}_2(\text{CO})_{10}]$, $[\text{Fe}(\text{CO})_2(\text{NO})_2]^0$,

$[\text{PtCl}_3(\eta^2\text{-C}_2\text{H}_4)]^\ominus$, $[\text{Ag}(\text{CN})_2]^{-1}$, $[\text{Fe}_2(\text{CO})_9]^0$

4. In the ionic reaction,
 $x\text{BrO}_3^- + y\text{Cr}^{3+} + z\text{H}_2\text{O} \rightarrow p\text{Br}_2 + q\text{HCrO}_4^- + r\text{H}^+$
 { x, y, z, p, q, r , are in simplest whole number ratio}
 Find the value of $(x + y + z)$

5. A 1 litre solution containing x millimoles each of FeO and $\text{Fe}_{0.8}\text{O}$ was completely titrated with 35 ml of 0.3M KMnO_4 in acidic medium. Find value of ' x '?

6. Gas A taken in a closed rigid container is allowed to decompose partially according to the reaction.

$\text{A(g)} \longrightarrow 2\text{B(g)} + 3\text{C(g)}$. The gaseous mixture formed effuses 1.5 times faster than a gas having molecular weight 103.5 amu under similar condition. If the mole fraction of C in the final gaseous mixture after reaction is ' X ', then the value of ' $10X$ ' is -

Given : Mol. wt. of A = 138

Mol. wt. of B = 64

Mol. wt. of C = 4

7. The graph of compressibility factor (Z) vs P for one mole of a real gas is plotted at constant temperature 273 K. If the slope of graph at very high pressure $\left(\frac{dZ}{dP}\right)$ is $\frac{1}{10} \text{ atm}^{-1}$, the volume of one molecule of real gas in cm^3 is $x \times 10^{-22}$. Find ' x '. [Take $N_A = 6 \times 10^{23}$, $R = 0.0821 \text{ lit atm/K mole}$]

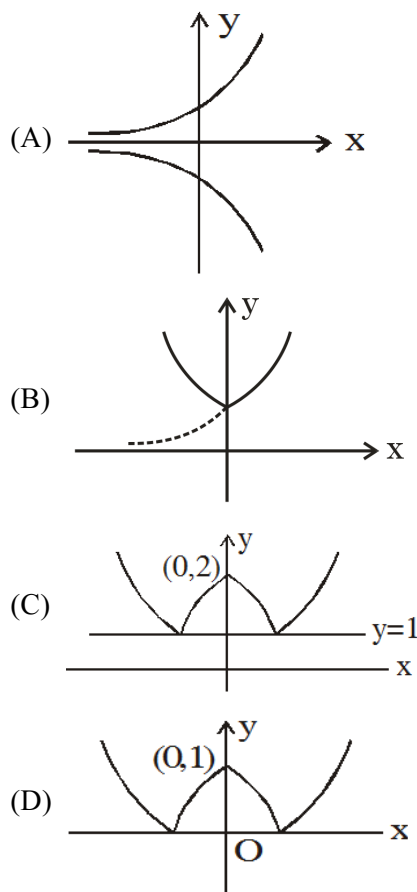
MATHEMATICS

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

- The system of equations
 $(1 - \lambda)x + 3y - 4z = 0$
 $x - (3 + \lambda)y + 5z = 0$
 $3x + y - \lambda z = 0$
 possesses non-trivial solutions for
 (A) exactly three distinct real values of λ .
 (B) exactly two distinct real values of λ .
 (C) exactly one real value of λ .
 (D) infinite number of real values of λ .
- If $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c}$ and side $a = 2$ then area of triangle is
 (A) $\sqrt{3}$ (B) 4
 (C) 3 (D) $\frac{\sqrt{3}}{4}$
- The function $f(x) = 2 + 4x^2 + 6x^4 + 8x^6$ has :
 (A) only one maxima
 (B) only one minima
 (C) no maxima and minima
 (D) many maxima and minima

- The global maximum value of the function
 $f(x) = x^3 - \frac{15}{2}x^2 + 18x + 2020$ on the
 set $A = \{x \mid x^2 - 4x \leq 0\}$
 (A) 2034 (B) 2036
 (C) does not exist (D) 2156
- If the surface area of a sphere of radius r is increasing uniformly at the rate $16\text{cm}^2/\text{sec}$ then the rate of change of its volume is proportional to-
 (A) $\sqrt{r}$ (B) r^2 (C) r (D) $r^{3/2}$
- The graph of $y = |e^{|x|} - 2|$ is



7. If $A : \{1, 2, 3, 4\}$ then which of the following relations from $A \rightarrow A$ is a function

- (A) $\{(1, 4), (2, 6), (1, 5), (3, 9)\}$
 (B) $\{(3, 3), (2, 1), (1, 2), (2, 3)\}$
 (C) $\{(1, 2), (2, 2), (3, 2), (4, 2)\}$
 (D) $\{(3, 1), (3, 2), (3, 3), (3, 4)\}$

8. The correct order is

- (A) $\sec^{-1}(-4) < \sec^{-1}(-3) < \sec^{-1}(3) < \sec^{-1}(4)$
 (B) $\sec^{-1}(-4) > \sec^{-1}(-3) > \sec^{-1}(3) > \sec^{-1}(4)$
 (C) $\sec^{-1}(-3) > \sec^{-1}(-4) > \sec^{-1}(4) > \sec^{-1}(3)$
 (D) $\sec^{-1}(-4) > \sec^{-1}(-3) > \sec^{-1}(4) > \sec^{-1}(3)$

9. $\int \frac{x^7}{(2x^2 + 1)^5} dx$ is equal to-
 (where C is constant of integration)

- (A) $\frac{1}{8x} \left(2x + \frac{1}{x}\right)^4 + C$
 (B) $\frac{1}{8} \frac{x^8}{(2x^2 + 1)^4} + C$
 (C) $\frac{1}{4} (2x^2 + 1)^{-4} + C$
 (D) $\frac{1}{4} \frac{(2x^2 + 1)^{-4}}{x} + C$

10. $\int e^x (1 - \cot x + \cot^2 x) dx =$

- (A) $e^x \cot x + c$
 (B) $-e^x \cot x + c$
 (C) $e^x \operatorname{cosec} x + c$
 (D) $-e^x \operatorname{cosec} x + c$

11. $\lim_{x \rightarrow \infty} \left(\frac{x+2}{x-2} \right)^{x+1}$ is equal to

- (A) e^4 (B) e^{-4}
 (C) e^2 (D) none of these

12. If $f(x)$ is continuous and differentiable over $[-2, 5]$ and $-4 \leq f'(x) \leq 3$ for all x in $(-2, 5)$, then the greatest possible value of $f(5) - f(-2)$ is -

- (A) 7 (B) 9 (C) 15 (D) 21

13. If $f(x) = \begin{cases} x^2 - ax + 3, & x \text{ is rational} \\ 2 - x, & x \text{ is irrational} \end{cases}$ is

continuous at exactly two points, then the range of possible values of a is

- (A) $(2, \infty)$
 (B) $(-\infty, 3)$
 (C) $(-\infty, -1) \cup (3, \infty)$
 (D) None of these

14. Let $f(x)$ is a polynomial function such that $f(x) + f\left(\frac{1}{x}\right) = f(x) \cdot f\left(\frac{1}{x}\right) \forall x \in \mathbb{R} - \{0\}$ and $f(5) = 126$, then value of $\int f(x) dx$ is-

- (A) $\frac{x^4}{4} + x + C$
 (B) $\frac{x^4}{4} - x + C$
 (C) $-\frac{x^4}{4} + x - C$
 (D) $-\frac{x^4}{4} - x - C$

(where C is an integration constant)

15. In a ΔABC , $\sin A$ and $\sin B$ are roots of $c^2x^2 - c(a+b)x + ab = 0$, where a, b, c are sides of triangle then the triangle is :-

(A) Equilateral
(B) Isosceles
(C) Right angled
(D) Acute angled

16. $\int \frac{(1 - \tan^2 x) \cdot \operatorname{cosec} 2x dx}{\sec^2 x}$ is equal to
[Note: 'C' is constant of integration]

(A) $\frac{\ln |\sin 2x|}{2} + C$
(B) $\frac{\ln |\cos 2x|}{2} + C$
(C) $\frac{\ln(\tan x)}{2} + C$
(D) $\frac{\ln |\cos x + \sin x|}{2} + C$

17. If $f(x) = \frac{\sin \frac{\pi x}{4}}{x+1}$, then $\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h^2 + 2h}$ is -

(A) $\frac{\pi - 4}{2\sqrt{2}}$
(B) $\frac{\pi}{16\sqrt{2}}$
(C) $\frac{\pi - 2}{8\sqrt{2}}$
(D) $\frac{\pi - 2}{16\sqrt{2}}$

18. If $f(x) = \int_2^{\frac{\sin x}{\log_e 2}} \left(\frac{5 \cos x + \cos^3 x}{(2 - \sin x) \sqrt{3 + \cos^2 x}} \right) dx$

and $f(0) = 1$ then value of $f(\pi/2)$ equals

(A) $\sqrt{3}e$
(B) $\frac{\sqrt{3}e}{2}$
(C) $\frac{3e}{2}$
(D) $3e$

19. The point on the curve $y^3 = 4x^2 + 12y$ at which the tangent is horizontal is

(A) (2,2)
(B) (-2,-2)
(C) (0,0)
(D) (1,1)

20. $\int \frac{\sqrt{x-1}}{x\sqrt{x+1}} dx$ is equal to

(A) $\ln |x - \sqrt{x^2 - 1}| - \tan^{-1} x + c$
(B) $\ln |x + \sqrt{x^2 - 1}| - \tan^{-1} x + c$
(C) $\ln |x - \sqrt{x^2 - 1}| - \sec^{-1} x + c$
(D) $\ln |x + \sqrt{x^2 - 1}| - \sec^{-1} x + c$
(c is constant of integration)

SEC-B : (Maximum Marks: 40)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type question.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. If $f(x) = x^3 + x$ then least integral value of x for which $f^3(x) + f(x) - f(4 - 2x) > 0$ is
2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by
$$f(x) = \begin{cases} k - x, & x \leq -1 \\ x^2 + 3, & x > -1 \end{cases}$$
. If $f(x)$ has least value at $x = -1$, then the greatest possible value of k is
3. If a derivable function $y = f(x)$ satisfies the condition $f(x) + f(y) = f(x)f(y) + f(xy) \forall x, y \in \mathbb{R}$ where $f(1) = 0$ and $f'(1) = -2$, then the number of solution(s) of the equation $f(x) = 2$

4. If a & b are real numbers such that $ab = 1$,
$$\Delta = \begin{vmatrix} 1 + a^2 - b^2 & 2ab & -2b \\ 2ab & 1 - a^2 + b^2 & 2a \\ 2b & -2a & 1 - a^2 - b^2 \end{vmatrix}$$
 then minimum value of Δ is
5. The sum of the infinite terms of the series $\cot^{-1} \frac{7}{4} + \cot^{-1} \frac{19}{4} + \cot^{-1} \frac{39}{4} + \cot^{-1} \frac{67}{4} + \dots$ is equal to $\tan^{-1}(K)$, then K is
6. The value of $\int_0^1 \sqrt[3]{2x^3 - 3x^2 - x + 1} \, dx$ is -
7. If $P(h, h^3 + 1)$, $h \in \mathbb{R}$ is situated at a least distance from the curve $(x - 4)^2 + (y - 1)^2 = 1$ then $3h^5 + h$ is equal to
8. In $\triangle ABC$, if $27\Delta^2 = s^4$ and $R = 4$ then area of $\triangle ABC$ (symbols used have usual notation in triangle ABC)
9. $\int_{1/5}^5 \frac{1}{x} 2^{x^2 + \frac{1}{x^2}} \left(x^3 - \frac{1}{x^3} \right) dx$ is :-
10. Number of point(s) of inflection of $f(x) = x^4 + 6x^3 + 35x^2 + 6x - 3$ is -

Space for Rough Work